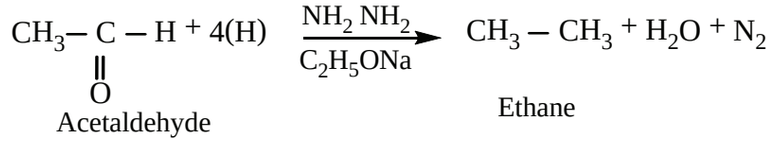
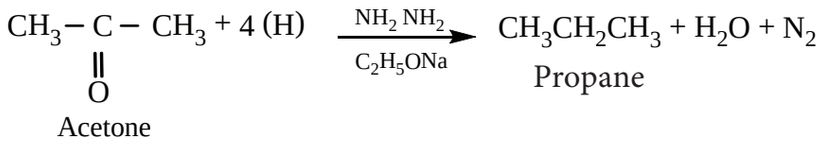
b) Wolf Kishner reduction Aldehydes and Ketones when heated with hydrazine (NH_2NH_2) and sodium ethoxide, hydrocarbons are formed Hydrazine acts as a reducing agent and sodium ethoxide as a catalyst.

Example



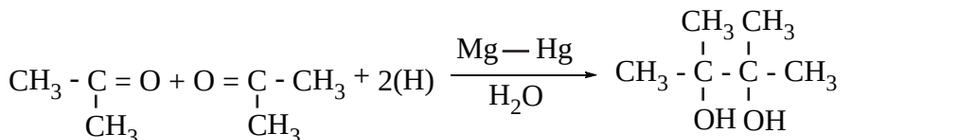
Acetaldehyde Ethane



Acetone Propane

Aldehyde (or) ketones is first converted to its hydrazone which on heating with strong base gives hydrocarbons.

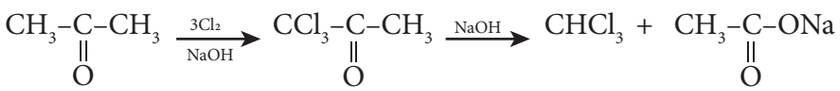
(iii) Reduction to pinacols: Ketones, on reduction with magnesium amalgam and water, are reduced to symmetrical diols known as pinacol.



Acetone Acetone 2,3 dimethyl butane 2,3 - diol (pinacol)

D) Haloform reaction

Acetaldehyde and methyl ketones, containing $CH_3-\overset{\overset{O}{\|}}{C}-$ group, when treated with halogen and alkali give the corresponding haloform. This is known as Haloform reaction.



E) Reaction involving alkylgroup

i) Aldol condensation

The carbon attached to carbonyl carbon is called α - **carbon** and the hydrogen atom attached to α - carbon is called α - hydrogen.

In presence of dilute base NaOH, or KOH, two molecules of an aldehyde or ketone having α - hydrogen add together to give β- hydroxyl aldehyde (aldol) or β - hydroxyl ketone (ketol). The reaction is called **aldol condensation reaction**. The aldol or ketol readily loses water to give α,β – unsaturated compounds which are aldol condensation products.

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